

Status

RK3588 / Orange Pi 5 Max — A/B-virt Emulator — Status

Revision: 1 **Last modified:** 2026-06-11T12:10:00Z **Scope:** The hardware-free emulation ladder that exercises the Helix OTA native Android A/B update flow (update_engine + AVB/dm-verity + auto-rollback) for RK3588 / Orange Pi 5 Max targets, on a developer host with NO live board. **Authority:** §11.4.45 (integration Status doc), §11.4.5 (captured-evidence table), §11.4.6 (no-guessing — pending work is stated as PENDING, never implied done), §11.4.107 (liveness, not a single frame), §11.4.112 (structurally-gated tiers SKIP honestly). **HEAD at last update:** 4278aa9.

Operator-blocked / Pending items (read first — §11.4.45(9))

Item	State	What is needed	§-ref
A/B slot switch + dm-verity + auto-rollback (the real OTA-apply flow)	PENDING — in progress	NOT yet proven. The base guest boots; the U-Boot bootcount + RAUC dm-verity A/B-slot layers are being built into the image (build7 in flight) and have NO captured slot-switch / rollback evidence yet.	§11.4.6
Tier-2 Cuttlefish — REAL Android A/B OTA	SKIP on this host	No /dev/kvm on this Apple-Silicon macOS dev host (confirmed absent). Ready to RUN on the operator's incoming Linux +	§11.4.3 / §11.4.112

Item	State	What is needed	§-ref
		nested-KVM host. NOT a fake PASS.	
Tier-3 — real RK3588 / Orange Pi 5 Max hardware	PENDING — no board	Requires physical hardware on the bench.	§11.4.6

Fidelity ladder (honest)

The emulation tiers trade fidelity for hardware-freedom. Higher tiers exercise more of the real OTA stack but demand scarcer environments.

Tier	What it proves	Fidelity	State
T0 — Containerized control-plane + device emulator (podman)	Control-plane $\Leftrightarrow$ device round-trip (register $\rightarrow$ update-check $\rightarrow$ telemetry) with NO live hardware	Protocol / control- plane only — no real A/B apply	SHIPPED
T1 — A/B-virt base image on QEMU virt + HVF	The aarch64 guest boots to a live interactive Linux userspace on the Apple CPU — the foundation the A/ B layers build on	Real kernel + userspace boot; A/ B-apply layers NOT yet present	FOUNDATION GREEN (boot); A/B layers in progress
T2 — Cuttlefish (cvd) on Linux + nested KVM	The real Android update_engine A/B + AVB/dm- verity + auto- rollback apply flow	Closest hardware- free proxy for the RK3588 OTA apply	SKIP on this host (no /dev/kvm); ready for Linux host
T3 — Real RK3588 / Orange Pi 5 Max board	The genuine on- device OTA apply + AVB/dm-verity + bootloader rollback	Full fidelity	PENDING — no hardware

Captured-evidence status table (§11.4.5)

Closed verdict vocabulary: PASS / FAIL / SKIP / PENDING_FORENSICS / OPERATOR-BLOCKED. Every PASS/SKIP cites a real evidence path verified to exist in this repo at revision time.

#	Capability under test	Verdict	Evidence (real, repo-relative)	Notes
E1	PWU-AB-0/1 base guest image built (aarch64 Buildroot, kernel 6.1.44 + ext2 rootfs)	PASS	tests/emulator/ab_virt/out/images/Image (MD5 9f3670cd7dba7bdeeebe2c6d791e929a), tests/emulator/ab_virt/out/images/rootfs.ext2 (MD5 a056760e88eea575977be13e38cfe430); builder tests/emulator/ab_virt/build_image.sh	The com PWU artif Boobu is in will them reco artif at re
E2	PWU-AB-1 FOUNDATION — boots to LIVE interactive userspace on QEMU virt + HVF	PASS	docs/qa/20260611T061626Z-ab-virt-boot/console.log (196 lines) — kernel boots on Apple CPU (physical CPU 0x000000000000 [0x610f0000], MIDR 0x610f), buildroot login: root, post-login uname -a → Linux buildroot 6.1.44 ... aarch64, sentinel HELIX_USERSPACE_LIVE_OK (line 182, emitted only from an interactive shell after login), clean poweroff. Driver mirror: .../console.log.driver.	§11. liver boot post sent sing Driv emu ab_ boo
E3	A/B slot switch (slot A→B select)	PENDING_FORENSICS	— (no captured slot-switch evidence yet)	NO Boo + R. laye built ima, Hon per no f
E4	dm-verity integrity on the booted slot	PENDING_FORENSICS	—	Dep Not exer
E5	Auto-rollback on failed boot (bootcount → fall back to known-good slot)	PENDING_FORENSICS	—	Dep E4. exer
E6	T0 containerized control-plane ⇔ device round-trip	PASS	Driver tests/emulator/tier1_container_e2e.sh (boots ota-server + ota-device-emu in a podman pod; asserts register → update-check → telemetry from captured container logs under docs/qa/<run-id>/)	Prot plan only NO real
E7	T2 Cuttlefish — REAL	SKIP	Driver tests/emulator/tier2_cuttlefish_ab.sh; host gate: /	§11. §11. topo

#	Capability under test	Verdict	Evidence (real, repo-relative)	Notes
	Android A/B OTA apply		dev/kvm absent on this Apple-Silicon macOS host (verified)	(existing fake script head exact invocation UNCONFIRMED pending Linux run. the outcome is inconsistent with host)
E8	T3 real RK3588 / Orange Pi 5 Max hardware	PENDING_FORENSICS	—	No benefit

What is genuinely proven today (§11.4.6)

- **Proven (captured evidence):** the A/B-virt base guest image **builds** and **boots to a live, interactive Linux userspace** on QEMU virt + HVF on this Apple-Silicon host (E1, E2) — the foundation tier.
- **NOT proven / honestly pending:** the real A/B slot switch, dm-verity, and auto-rollback flow (E3–E5) — the U-Boot+RAUC layers are still being built; there is no slot-switch or rollback evidence yet.
- **Honestly skipped (topology-gated):** the Tier-2 Cuttlefish real-Android-A/B path (E7) SKIPS on this host for lack of /dev/kvm and is ready to run on a Linux + nested-KVM host. No fabricated continuity, no fake PASS.

Provenance

Commit	Subject
dd43738	research + dev-host A/B-virt build infra (base image build in progress)
d5374d0	PWU-AB-1 foundation GREEN — base image boots to LIVE userspace on QEMU+HVF
4278aa9	add U-Boot qemu_arm64 + RAUC + GPT tooling to the A/B-virt build (PWU-AB-1 full, in progress)